GUIDE FOR SAT EXAM

Thermodynamics and KTG Questions for SAT Exam

VSAQ

Very Short Answer Questions · 2 Marks

1 Write the sign convention for heat energy (Q) and work (W).

THERMODYNAMICS

Sol: In thermodynamics, the following sign convention is used:

i) Heat energy $Q > 0$: Heat is supplied to (absorbed by) the system.
$Q < 0$: Heat is released by the system to the surroundings.

ii) Work done : $W > 0$: Work is done by the system on the surroundings.
$W < 0$: Work is done on the system by the surroundings.

2 Write the specific heat relation for an ideal gas.

THERMODYNAMICS

A: For an ideal gas, the molar specific heat capacities at constant pressure (C_p) and at constant volume (C_v) are related by: $C_p - C_v = R$, R= Universal gas constant

3 Explain the Isochoric and Isobaric processes.

THERMODYNAMICS

A: 1) **Isochoric process** is the process that takes place at constant volume of the system.
Ex: Heat is added to the gas enclosed in a cylinder having rigid walls and a fixed piston.

2) **Isobaric process** is the process that takes place at constant pressure of the system.
Ex: Heating of water at atmospheric pressure.

4 Why a heat engine with 100% efficiency can never be realized in practice?

THERMODYNAMICS

A: The efficiency of heat engine is given by $\eta = 1 - \frac{T_2}{T_1}$. Thus η is always less than one.
The value of η can be 1 only if $T_2 = 0$. This is possible only if the sink is at absolute zero temperature which cannot be attained in reality.
So heat engine with 100% efficiency is not possible.

5 When does a real gas behave like an ideal gas?

KTG TS 15,16,18,19,22,23,24,25

A: At **low pressure** and **high temperature,** a real gas behaves like an ideal gas.

6 State Boyle's law and Charles law.

KTG

A: Boyle's law: At constant temperature(T), the Pressure (P) of a given mass of gas is inversely proportional to its Volume (V). $P \propto \frac{1}{V}$, (T is constant) (or) $PV =$ Constant,

Charles law: At constant pressure(P), the volume(V) of a given mass of a gas is directly proportional to its temperature(T) on Kelvin scale.

$V \propto T$, (P is constant) (or) $\frac{V}{T} =$ Constant

7 Define rms speed of gas molecules.

KTG

A: RMS Speed is the square root of the mean of the squares of the speeds of all molecules of a gas.

Formula: RMS Speed : $C = \sqrt{\frac{C_1^2 + C_2^2 + \cdots + C_n^2}{n}}$ (or) $C = \sqrt{\frac{3RT}{M}}$

8 The absolute temperature of a gas is increased 3 times. What will be the increase in rms velocity of the gas molecule?

KTG TS 15,19,20,25

A: RMS velocity of a gas molecule $C = \sqrt{\frac{3RT}{M}} \Rightarrow C \propto \sqrt{T} \Rightarrow \frac{C_1}{C_2} = \sqrt{\frac{T_1}{T_2}}$

Given $T_2 = 3\,T_1$

$\therefore \frac{C_1}{C_2} = \sqrt{\frac{T_1}{3T_1}} = \frac{1}{\sqrt{3}} \Rightarrow C_2 = \sqrt{3}\,C_1$

The r.m.s velocity of the gas molecule increases by $\sqrt{3}$ times of initial rms velocity.

SAQ Short Answer Questions

9 State and explain first law of thermodynamics.

THERMODYNAMICS TS 22

A: First law of Thermodynamics:

Statement: The first law of thermodynamics states that the amount of heat supplied to a system is equal to the sum of the change in internal energy of the system and the amount of external work done by the system.

If a small quantity of heat (dQ) is supplied to a gas , a part of it is used to increase internal energy(dU) and remaining heat is used in doing external work (dW).

Then $dQ = dU + dW$

Sign convention:

If the work is done **by** the system, dW is **positive**

If the work is done **on** the system, dW is **negative**

When heat is **supplied** to the system, dQ is **positive**

When the heat is **taken out** from the system, dQ is **negative**

The essential content of the first law of thermodynamics is that it defines one of the thermodynamic quantities 'the internal energy' or 'internal energy function'.

This law is a consequence of conservation of energy.

Limitations:

1) It does not tell about the direction of heat flow.

2) It does not tell us about the efficiency with which heat can be converted into work.

10 Compare isothermal and an adiabatic processes.

THERMODYNAMICS TS 22

A:

Isothermal Process	Adiabatic Process
1) In Isothermal process, temperature remains constant and heat exchanges between system and surroundings.	1) In Adiabatic process, total heat remains constant and temperature changes between system and surroundings.
2) This process should be performed in a good conducting vessel.	2) This process should be performed in a bad conducting vessel.
3) It is a slow process.	3) It is a quick process
4) Gas law $PV =$ constant,holds true.	4) Gas law $PV^{\gamma} =$ constant, holds true.
5) Specific heat is infinity.	5) Specific heat is zero.
6) Change in internal energy is zero($dU = 0$).	6) There is some external work done($dW \neq 0$)
7) First law of Thermodynamics becomes $dQ = dW$	7) First law of Thermodynamics becomes $dU + dW = 0\,(dQ = 0)$.
8) **Ex:** Boiling of water in open air.	8) **Ex:** Compressing an ideal gas quickly.

11 **Four molecules of a gas have speeds 1,2,3 and 4 km/s. Find the rms speed of the gas molecule.**

KTG BMP

Sol: The r.m.s speed of n gas molecules is $C = \sqrt{\frac{C_1^2 + C_2^2 + \cdots + C_n^2}{n}}$

Here, $n = 4$, $C_1 = 1$ km/s, $C_2 = 2$ km/s, $C_3 = 3$ km/s, $C_4 = 4$ km/s

$$\therefore C = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2}{4}} = \sqrt{\frac{1 + 4 + 9 + 16}{4}} = \sqrt{\frac{30}{4}} = \frac{\sqrt{30}}{2} = 2.74 \text{ km / s}$$

LAQ Long Answer Questions

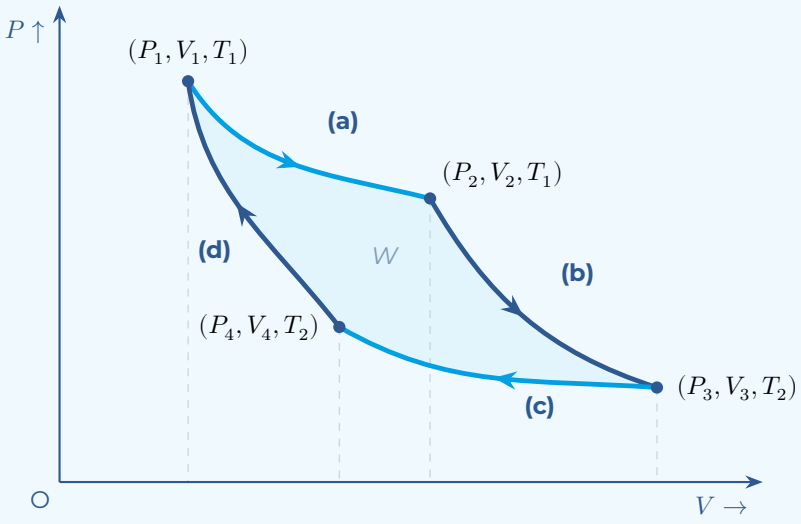
12 **Describe the working of Carnot engine and obtain an expression for its efficiency. A Carnot engine whose efficiency is 45% takes heat from a source maintained at a temperature of 227°C, then find the temperature of sink.**

THERMODYNAMICS TS 15,17,19,25

A: Carnot engine is a reversible heat engine operating between two temperatures.

Working of Carnot engine: The carnot engine undergoes a cycle of processes called carnot cycle. It consists of two isothermal processes connected by two adiabatic processes. Ideal gas acts as the working substance in the carnot engine.

The steps of Carnot cycle:



a) Isothermal expansion of the gas taking its state from (P_1, V_1, T_1) to (P_2, V_2, T_1).

Heat(Q_1) absorbed by the gas from the reservoir at temperature (T_1) is equal to the work done (W_1) by the gas.

$$W_1 = Q_1 = nRT_1 \log_e\left(\frac{V_2}{V_1}\right) \quad (1)$$

b) Adiabatic expansion of the gas from (P_2, V_2, T_1) to (P_3, V_3, T_2).

Work done by the gas is

$$W_2 = \frac{nR}{(\gamma-1)}(T_1 - T_2) \tag{2}$$

c) Isothermal compression of the gas from (P_3, V_3, T_2) to (P_4, V_4, T_2).

Heat(Q_2) released by the gas to the reservoir at temperature(T_2)is equal to the work done on the gas .

$$W_3 = Q_2 = nRT_2 \log_e\left(\frac{V_3}{V_4}\right) \tag{3}$$

d) Adiabatic compression of the gas from (P_4, V_4, T_2) to (P_1, V_1, T_1).

Work done on the gas is

$$W_4 = \frac{nR}{(\gamma-1)}(T_1 - T_2) \tag{4}$$

∴ Total work done by the gas in one complete cycle is $W = W_1 + W_2 - W_3 - W_4$

$$= nRT_1 \log_e\left(\frac{V_2}{V_1}\right) + \frac{nR}{\cancel{(\gamma-1)}}\cancel{(T_1 - T_2)} - nRT_2 \log_e\left(\frac{V_3}{V_4}\right) - \frac{nR}{\cancel{(\gamma-1)}}\cancel{(T_1 - T_2)}$$

$$W = nRT_1 \log_e\left(\frac{V_2}{V_1}\right) - nRT_2 \log_e\left(\frac{V_3}{V_4}\right) \tag{5}$$

The efficiency of the Carnot engine is

$$\eta = \frac{W}{Q_1} = \frac{\cancel{nR}T_1 \log_e\left(\frac{V_2}{V_1}\right) - \cancel{nR}T_2 \log_e\left(\frac{V_3}{V_4}\right)}{\cancel{nR}T_1 \log_e\left(\frac{V_2}{V_1}\right)}$$

$$\eta = \frac{T_1 \cancel{\log_e\left(\frac{V_2}{V_1}\right)}}{T_1 \cancel{\log_e\left(\frac{V_2}{V_1}\right)}} - \frac{T_2 \log_e\left(\frac{V_3}{V_4}\right)}{T_1 \log_e\left(\frac{V_2}{V_1}\right)} = 1 - \left(\frac{T_2}{T_1}\right)\frac{\log_e\left(\frac{V_3}{V_4}\right)}{\log_e\left(\frac{V_2}{V_1}\right)} \tag{6}$$

(b) & (d) are adiabatic processes.

$\therefore T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1}$ ($\because TV^{\gamma-1}$ is constant)

$\therefore T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$ ($\because TV^{\gamma-1}$ is constant)

Dividing the above two equations, we get

$$\left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{V_3}{V_4}\right)^{\gamma-1} \Rightarrow \frac{V_2}{V_1} = \frac{V_3}{V_4} \tag{7}$$

Substitute (7) in (6) we get $\eta = 1 - \left(\frac{T_2}{T_1}\right)\frac{\cancel{\log_e\left(\frac{V_2}{V_1}\right)}}{\cancel{\log_e\left(\frac{V_2}{V_1}\right)}}$

From equations (6) & (7), efficiency of the Carnot Engine is

$$\boxed{\eta = 1 - \frac{T_2}{T_1}} \tag{8}$$

Problem-Solution:

Given $\eta = 45\% = 0.45,\ T_1 = 227^\circ\text{C} + 273^\circ\text{C} = 500\,\text{K}$

Carnot engine's efficiency, $\eta = 1 - \frac{T_2}{T_1}$

$$\Rightarrow 0.45 = 1 - \frac{T_2}{500} \Rightarrow \frac{T_2}{500} = 1 - 0.45 = 0.55 \Rightarrow T_2 = 0.55 \times 500 = 275\,\text{K}$$

Converting to Celsius, $T_2 = 275 - 273 = 2^\circ\text{C}$

13 State second law of thermodynamics. How is heat engine different from a refrigerator. A heat engine takes in 1420 J of heat from the source and exhausts 660 J of heat to the sink — how much work is done by the engine?

THERMODYNAMICS TS 18, 20,23,24

A: Second law of thermodynamics:

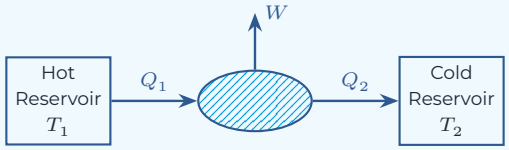
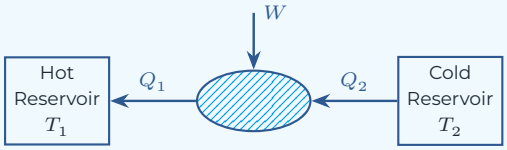
Second law of thermodynamics gives the direction of flow of heat.

It consists of two statements.

I) Kelvin - Planck Statement: It is impossible to construct a heat engine that absorbs heat from a hot reservoir which converts completely that heat into work .

(or) It is impossible to construct an ideal heat engine with 100% thermal efficiency.

II) Clausius Statement: It is impossible to transfer heat from a colder object to a hotter object without doing external work (or) It is impossible to construct an ideal refrigerator.

HEAT ENGINE	REFRIGERATOR
	
1) Heat engine converts heat into work.	1) Refrigerator is a reversed heat engine.
2) The working substance absorbs heat (Q_1) from the hot reservoir at high temperature (T_1)	2) The working substance absorbs heat (Q_2) from the cold reservoir at low temperature (T_2)
3) The working substance rejects heat (Q_2) to cold reservoir at lower temperature (T_2)	3) The working substance rejects heat (Q_1) to the hot reservoir at high temperature (T_1)
4) Here, work (W) is done by the system.	4) Here, external work is done on the system
5) The efficiency (η) of a heat engine is $\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$	5) The coefficient of performance of a refrigerator is $\alpha = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$
6) η is less than 1.	6) α is usually greater than 1.

Problem- Solution:

Given $Q_1 = 1420\,\text{J},\ Q_2 = 660\,\text{J}$

Formula: Work done $W = Q_1 - Q_2 = 1420 - 660 = 760\,\text{J}$